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San Jose

Poughkeepsie

Mr. Ray Haug

February 20th 1956

Project Stretch Requirements for the RAM

This is to confirm our telephone conversation of February 15, 1956. It is my understanding that we have agreed on the following:

1. Number of bits in parallel - 60 to 68. If a large increase in density can be obtained electronic de-skewing may be acceptable.
2. Disk size - 18" diameter
3. Storage capacity per 60 sides - 1,048,576
4. Number of tracks per side - 128
5. Number of bits per track - 8912
6. Number of blocks per track - 4
7. Separation between blocks - at least 4 bits

The above specifications indicate a bit density of roughly 205 to the inch and a track spacing of 50 to the inch. Obviously, increased bit density would permit fewer tracks and wider spacing between tracks. The above number of bits per track and the number of tracks per side are binary members. If a large increase in capacity can be obtained by making these no longer binary members, this should be considered.

In addition, I believe that we agreed that there would be two sets of heads and two actuator mechanisms per set of discs, i.e. per 60 to 68 sides. Other than the usual considerations of space, reliability and economy, the number of sets of discs per shaft is not important.

I presume that since the read and write circuits are intimately connected with the head design that you would want to develop these in cooperation with Joe Logue. The control logic and the synchronizing register will probably want to be circuits which will be similar to those in the Stretch machine. We will discuss these later. I am still assuming that the timing of the RAM will be accomplished by one or more timing tracks, perhaps using a single track and a fixed head.

I hope the above specifications and considerations are compatible with the commercial RAM design. It seems to me that they are sufficiently conservative so that they should be quite reliable and not unduly expensive for commercial use. Please let me know if any small modification would bring this close to the commercial RAM.

I enjoyed my visit to the San Jose Laboratory and I want to thank you for your cooperation, and especially, the weather.

W. P. Edwards

WPE:adm

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